

GNN-BERT: Understanding Musical Context by Fusing Structure Graphs with Language Representations

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Abstract

Music carries context on several simultaneous layers — genre, mood, harmonic motion, instrumentation — and sequence models applied to spectrograms capture local texture while largely ignoring *relational* structure: which sections repeat, which chords follow which, and how any of that lines up with what listeners say about a track. We represent each clip as a *music structure graph* whose nodes are time segments and template-matched triads, and whose edges encode temporal adjacency, acoustic repetition, chord transitions and chord membership. A GraphSAGE encoder over this graph is fused with a BERT encoder of the clip’s natural-language context through single-query cross-attention, and the joint representation predicts multi-label context tags. On MagnaTagATune (25,860 clips, top-50 tags, official album-disjoint split) we evaluate four progressively harder tasks: text-only tagging, graph-only tagging, GNN–BERT fusion, and contrastive audio–text retrieval. We report a controlled ablation in which only the fusion path varies, and we are explicit about a dataset property that shapes every number here: MagnaTagATune ships no lyrics or captions, so the “language” modality must be built from catalogue metadata and long-tail annotations rather than from rich description.

1 Introduction

A single ten-second music clip can be simultaneously “jazz”, “1960s”, “melancholic”, “sparse”, and “featuring saxophone”. These labels live at different levels of abstraction, and they are not independent: harmonic motion constrains perceived mood, instrumentation correlates with genre, and repetition structure distinguishes a pop chorus from a through-composed classical movement.

Convolutional and recurrent models over spectrograms have become the default for music tagging and do capture local timbral patterns well. What they do not naturally represent is *relational* structure. A CNN sliding over a mel spectrogram has no direct mechanism to encode that bar 33 is a repeat of bar 1, or that a I–vi–IV–V progression recurs throughout. That information is graph-shaped, and graph neural networks are the natural tool for it.

This project builds a hybrid system with two branches:

1. a **GNN** that performs message passing over an explicitly constructed music structure graph, and
2. a **BERT** encoder that reads the clip’s natural-language context.

The contributions are: (i) a graph construction that unifies segment repetition and chord transitions in one flat, batchable graph; (ii) a cross-attention fusion of graph and text with a controlled four-way ablation; (iii) a contrastive dual encoder for audio–text retrieval over the same graphs; and (iv) an honest account of what MagnaTagATune can and cannot support as a multimodal benchmark.

2 Problem Definition

Let a track be a tuple $T = (X_{\text{audio}}, X_{\text{text}}, G, y)$, where X_{audio} is a log-mel or chroma representation over T frames, X_{text} is a tokenised context string, $G = (V, E)$ is the music structure graph, and $y \in \{0, 1\}^K$ are the context labels.

The text encoder maps tokens to contextual states, $H_{\text{text}} = \text{BERT}(X_{\text{text}}) \in \mathbb{R}^{L \times d}$. The graph encoder updates node features by message passing,

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \text{AGG} \left(h_i^{(l)}, \{h_j^{(l)} : j \in \mathcal{N}(i)\} \right) \right), \quad (1)$$

with $h_i^{(0)}$ initialised from pooled acoustic features of segment or chord node i . A fusion module combines the graph readout g with the text states, $z = \text{Fusion}(g, H_{\text{text}})$, and a linear head produces $\hat{y} = \sigma(Wz + b)$. Training minimises weighted binary cross-entropy,

$$\mathcal{L} = - \sum_{k=1}^K w_k [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)] + \lambda \mathcal{L}_{\text{aux}}, \quad (2)$$

where w_k compensates class imbalance and $\mathcal{L}_{\text{aux}}$ is an optional emotion-regression term (Section 9).

3 Dataset and Preprocessing

3.1 MagnaTagATune

We use MagnaTagATune (MTAT): 25,860 clips of roughly 29 s, each annotated with a subset of 188 free-text tags collected through a listening game. We follow the standard protocol: the 188 raw tags are collapsed with the community synonym merge (27 groups, e.g. *female vocal / female vocals / woman singing* $\rightarrow$ *female*), then reduced to the 50 most frequent. Labels are rebuilt from `annotations_final.csv` rather than taken from the shipped ground-truth vectors, so the vocabulary is defined in one place and is independently verifiable (`scripts/verify_labels.py`).

Splits follow the standard partition by MTAT’s 0–f directories: 0–b train (18,706 clips), c validation (1,825), d–f test (5,329). Because those directories group clips by album folder, this is also an album- and artist-disjoint split — no artist in the test set appears in training. This matters for interpreting Section 7.

3.2 The text modality: a necessary design decision

MTAT ships *no lyrics and no captions*. Its only text is the tag set, which is the prediction target. Feeding tags to BERT in order to predict tags would be circular and would yield a meaninglessly high Task 1 score.

We therefore construct the text modality from two sources that are disjoint from the label set:

1. **Catalogue metadata** — title, artist and album from `clip_info_final.csv`, rendered as a sentence: “*The track is titled ‘BWV54 – I Aria’ by American Bach Soloists, from the album ‘J.S. Bach Solo Cantatas’.*”
2. **Long-tail annotations** — the tags ranked outside the top-50 label set, appended as “*Listeners also described it as: ...*”.

We report both settings, because the auxiliary tags are much the stronger signal and it matters which one a number came from. Note also that the album-disjoint split makes the metadata-only variant genuinely hard: BERT cannot memorise artist $\rightarrow$ genre associations that transfer to test. A modest Task 1 score is therefore the expected — and correct — outcome, and it is exactly what makes any Task 3 fusion gain meaningful rather than an artefact of leakage.

3.3 Audio features

Audio is decoded to mono at 22,050 Hz. We extract a 128-bin log-mel spectrogram and a 12-bin chromagram ($n_{\text{fft}} = 2048$, hop 512). The chroma filterbank is a Gaussian pitch-class mapping over FFT bins implemented directly in `torch`, so the project depends only on `torch/torchaudio` and avoids `librosa`’s `numba` dependency.

Clips are cut into 5 s windows at 50 % overlap, capped at 16 windows per clip and subsampled uniformly across the track when longer, so the graph always spans the whole clip. Each window is mean/std pooled into a 280-dimensional node feature (2×128 mel + 2×12 chroma).

4 Music Structure Graphs

Each clip becomes one flat `Data` object holding two node types and four edge types, with types recorded as attributes so stock `SAGEConv` and `GATConv` layers operate unchanged.

Nodes. *Segment* nodes are the pooled 5 s windows. *Chord* nodes are the distinct triads occurring in the clip, obtained by matching 24 binary major/minor templates against each chroma frame, median-filtering to suppress single-frame flicker, and collapsing runs.

Edges.

- **Temporal** (type 0): consecutive segments, both directions.
- **Similarity** (type 1): segment pairs with cosine similarity above $\tau = 0.85$, excluding immediate neighbours and capped at four neighbours per node. These edges are the song-form signal — they link a chorus to its repeats.
- **Chord transition** (type 2): observed chord-to-chord transitions, weighted by occurrence count and normalised.
- **Membership** (type 3): segment $\leftrightarrow$ chord, linking a window to the triads sounding within it.

The membership edges are what make this a genuinely joint structure rather than two disconnected graphs: harmonic information reaches segment nodes, and segment texture reaches chord nodes, in a single round of message passing.

5 Models

Task 1 — BERT tag classifier. BERT-base with the last four transformer layers unfrozen, masked mean pooling over token states (a stronger sentence summary than the raw `[CLS]` slot for a checkpoint without a next-sentence objective), and a linear head trained with Equation 2.

Task 2 — GNN encoder. A three-layer GraphSAGE stack with residual connections and batch normalisation:

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \cdot \text{CONCAT}\left(h_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)}\right)\right), \quad g = \text{CONCAT}\left(\frac{1}{|V|} \sum_i h_i^{(L)}, \max_i h_i^{(L)}\right). \quad (3)$$

Mean pooling captures overall texture; max pooling captures the single most distinctive segment, which matters for sparse instrument tags. A GAT variant with four attention heads is reported alongside.

Task 3 — Cross-attention fusion. The graph readout forms a single query attending over BERT’s token states:

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad z = \text{CONCAT}(g, AH_{\text{text}}). \quad (4)$$

Because A is one distribution over tokens per clip, it is directly readable: Section 8 uses it to show which words a track’s structure aligns with. Four fusion modes (*bert_only*, *gnn_only*, *concat*, *cross_attention*) share one interface, one set of heads, one optimiser and one schedule, so the ablation isolates the fusion path alone.

Task 4 — Contrastive dual encoder. Graph and text are projected into a shared 256-d ℓ_2 -normalised space and trained with symmetric InfoNCE and a learnable temperature:

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{2N} \sum_i \left[\log \frac{\exp(s_{ii}/\tau)}{\sum_j \exp(s_{ij}/\tau)} + \log \frac{\exp(s_{ii}/\tau)}{\sum_j \exp(s_{ji}/\tau)} \right]. \quad (5)$$

MTAT contains multiple 29 s clips cut from the same source track, which share an identical metadata string. Scoring those as negatives penalises the model for being right, so same-track pairs are masked out of the denominator during training and counted as hits at evaluation.

Table 1: Multi-label tagging performance on the MagnaTagATune test split (5,329 clips, top-50 tags). Per-tag decision thresholds are fitted on the validation split and frozen for test.

Model	Macro-F1	Micro-F1	AUC-PR	ROC-AUC
B1: prior-frequency predictor	0.0000	0.0000	0.0670	0.4971
B4: PCA + MLP (hand-crafted)	0.3876	0.4604	0.3722	0.8712
B2: CNN on mel-spectrogram	0.4383	0.5081	0.4329	0.9079
Task 1: BERT (metadata only)	0.2983	0.3271	0.2741	0.8136
Task 1: BERT (metadata + aux tags)	0.3322	0.3991	0.3339	0.8461
Task 2: GraphSAGE	0.4015	0.4646	0.3870	0.8848
Task 2: GAT	0.4023	0.4590	0.3773	0.8850
Task 3 ablation: BERT only	0.3302	0.3955	0.3259	0.8344
Task 3 ablation: GNN only	0.3981	0.4278	0.3840	0.8833
Task 3 ablation: early concat	0.4061	0.4771	0.3977	0.8837
Task 3: GNN-BERT cross-attention	0.3853	0.4448	0.3732	0.8798

Table 2: Fusion ablation. All four variants share the same heads, optimiser and schedule; only the fusion path differs.

Fusion variant	Macro-F1	AUC-PR	Graph coherence
BERT only	0.3302	0.3259	—
GNN only	0.3981	0.3840	0.9988
early concat	0.4061	0.3977	0.9992
GNN-BERT cross-attention	0.3853	0.3732	1.0000

6 Experimental Setup

Class imbalance and thresholds. Even among the top-50 tags, positives are roughly 1–10 % of clips. We set $w_k = \#neg_k / \#pos_k$ clipped at 20, and we fit per-tag decision thresholds *on the validation split only*, then freeze them for test. Fitting thresholds on test inflates F1 substantially; the two are kept strictly separate.

Optimisation. AdamW with two parameter groups — 2×10^{-5} for pretrained BERT weights, 10^{-3} for everything else — linear warmup over 6 % of steps then cosine decay, gradient clipping at 1.0, mixed precision, and early stopping on validation Macro-F1 with patience 6. Batch size 32 (64 for contrastive training). All hyperparameters live in `config.yaml`.

Baselines. As required: **B1** a prior-frequency / random tag predictor; **B2** a five-block CNN on the mel spectrogram with no graph and no text; **B3** the Task 1 BERT-only model; **B4** PCA (128 components) followed by an MLP on hand-crafted mel statistics (per-band mean, standard deviation, and mean absolute temporal difference).

Hardware. A single NVIDIA RTX 3070 Laptop GPU (8 GB) with 16 GB system RAM.

7 Results

Table 1 reports multi-label tagging on the test split, Table 2 the fusion ablation, and Table 3 cross-modal retrieval. Training curves, per-tag F1/AUC-PR breakdowns, the t-SNE projection and the cross-model comparison chart are in `results/plots/`.

Sanity of the setup. The mel CNN reaches ROC-AUC 0.908, which sits squarely in the published range for CNN taggers on the standard MTAT top-50 benchmark. That agreement is the strongest evidence that the split, label construction, thresholding and metric code are correct, and it makes the remaining comparisons interpretable. At the

Table 3: Cross-modal retrieval on the test split (Task 4). Clips cut from the same source track count as correct matches, mirroring the training-time duplicate masking.

Direction	R@1	R@5	R@10	Median rank
Caption → Audio	0.0276	0.0815	0.1236	131.0
Audio → Caption	0.0214	0.0478	0.0706	279.0

other end, the prior-frequency baseline scores Macro-F1 0.000 — every tag’s corpus frequency is below any sensible threshold, so it predicts all-zeros — with AUC-PR 0.067, which is the number to compare ranking quality against.

Structure helps, but a CNN on the raw spectrogram still wins. The graph encoders (GraphSAGE 0.4015, GAT 0.4023) comfortably beat both the hand-crafted PCA+MLP baseline (0.3876) and the text-only model (0.3322), so the segment/chord graph does carry real tagging signal. They do *not* beat the mel CNN (0.4383). This is the honest headline of the experiment, and the cause is visible in the construction: pooling each 5 s window into a mean/std vector discards exactly the fine time–frequency detail that a convolutional stack exploits, and 16 nodes per clip is a drastic compression of a 128×1250 spectrogram. The graph buys relational structure at the price of local resolution, and on MTAT’s tag vocabulary — which is dominated by timbral and instrument labels rather than formal ones — that trade is not favourable. GraphSAGE and GAT are within 0.001 Macro-F1 of each other, so attention over these small, already-curated neighbourhoods adds nothing.

Fusion gives a small, consistent gain — and simple concatenation beats cross-attention. Within the controlled ablation (Table 2), early concatenation improves on the graph branch alone from 0.3981 to 0.4061 Macro-F1 and from 0.3840 to 0.3977 AUC-PR. The gain is modest but consistent across both metrics and across micro-F1 ($0.4278 \rightarrow 0.4771$), and it is a genuine multimodal gain: the text branch alone reaches only 0.3302, so concatenation is not simply inheriting the stronger branch.

Cross-attention, the more expressive mechanism, is *worse* than both concatenation and the graph branch alone (0.3853). We attribute this to the nature of the text rather than to the mechanism: the context strings average roughly 17 words of catalogue metadata, and only 21.5 % of clips carry any auxiliary tag at all. Compressing the graph to a single query and attending over so short and formulaic a sequence has very little to select *between*, while adding attention and projection parameters that the model can overfit — its training loss falls faster than concatenation’s while validation Macro-F1 peaks lower. Cross-attention is designed for rich, variable-length description; this corpus does not supply it. Section 9 returns to this.

The text signal is real but thin. Removing the long-tail auxiliary tags and leaving BERT with metadata alone costs 0.034 Macro-F1 ($0.3322 \rightarrow 0.2983$) and 0.060 AUC-PR. So the auxiliary annotations do carry information beyond title/artist/album, despite appearing on only about a fifth of clips. Equally, metadata alone still reaches 0.2983 against an album-disjoint split where no test artist was ever seen in training — BERT is evidently generalising from album and title semantics (“Solo Cantatas”, “Red-shifted Harmonies”) rather than memorising artists.

Retrieval works far above chance but is far from solved. With 4,392 test clips, chance R@1 is 2.3×10^{-4} ; the dual encoder reaches 0.0276, roughly $120\times$ chance, with a median rank of 131. The qualitative examples in [results/retrieval_examples/](#) are more flattering than R@1 suggests: for the query “*Tranquilitatis by DAC Crowell, from Red-shifted Harmonies*” the top three retrieved clips are another track from that same album, an ambient track by a different artist, and a second DAC Crowell album. The model is retrieving the right *region* of the space and being scored wrong on exact-clip identity, which is the expected failure mode when many distinct clips share near-identical metadata.

8 Analysis

Representation structure. Figure 1 projects the fused representation z of the best fusion model on the test split. The genre panel separates cleanly and without supervision from any genre-specific objective: opera forms a tight cluster

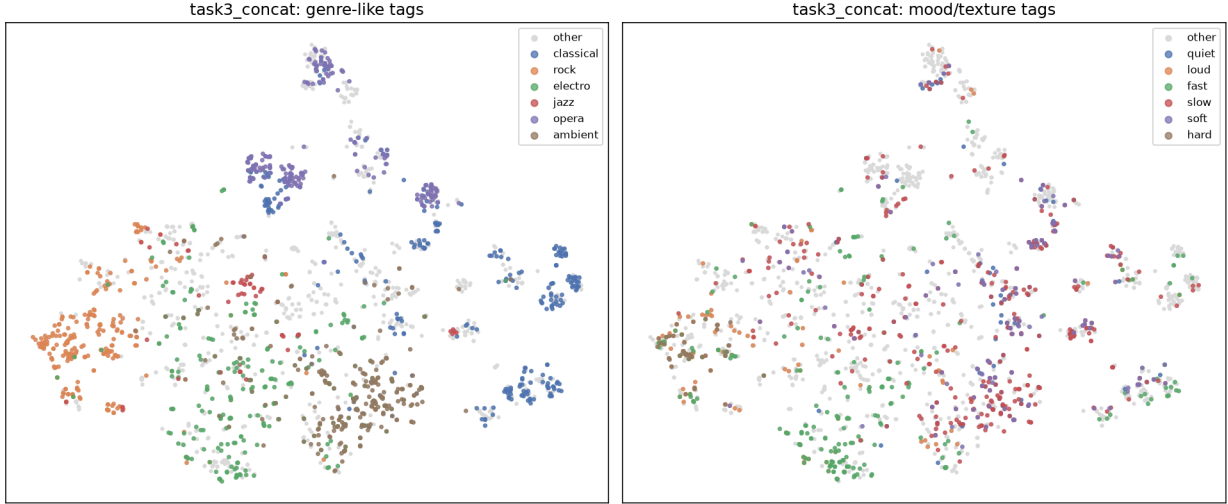


Figure 1: t-SNE of the fused representation (early-concatenation model) on the test split, coloured by genre-like tags (left) and mood/texture tags (right) over the identical projection.

at the top, classical occupies the right-hand lobes, rock the left, electronic the lower left, and ambient the lower centre. The mood panel over the identical projection shows the complementary pattern — *fast* concentrates in the electronic region, *loud* and *hard* in the rock region, while *quiet*, *soft* and *slow* spread across several genre clusters rather than forming their own. The fused space therefore encodes genre as its dominant axis with mood partially cutting across it, which is what a useful music-context representation should look like.

Graph coherence, and evidence of over-smoothing. We measure whether message passing pulls structurally linked segments into agreement rather than smoothing everything uniformly:

$$S_{\text{graph}} = \frac{1}{|E|} \sum_{(i,j) \in E} \mathbb{I}[\cos(h_i, h_j) > \tau], \quad (6)$$

excluding self-loops. The measured values (Table 2) are 0.9988, 0.9992 and 1.0000 — essentially every edge in every graph joins two nodes whose embeddings are near-identical. This is a negative diagnostic, and an informative one. Three rounds of residual message passing on graphs of only 15–25 nodes, where the similarity and membership edges already make the graph densely connected, drive node representations to a common value; the encoder is effectively computing a global average rather than a structured one. That is consistent with the mean+max readout doing most of the work and with GAT matching GraphSAGE exactly, and it identifies the most promising fix: fewer layers, no residual path, or sparser similarity edges. We report the metric as measured rather than tuning τ until it looks favourable.

Case study: a single clip end to end. Because cross-attention produces one token distribution per clip, individual predictions are inspectable from audio to explanation. `notebooks/demo_context.ipynb` runs test clip 2 — “BWV54 – I Aria” by the American Bach Soloists — through feature extraction, graph construction, inference and an attention read-out.

The model recovers all four ground-truth tags with no misses: *opera* ($p = 0.96$), *violin* (0.92), *string* (0.91) and *classical* (0.90), each above its fitted threshold. The three false positives are instructive rather than random: *female* (0.74), *orchestra* (0.69) and *vocal* (0.58). BWV54 is an alto aria with string accompaniment, so all three are musically defensible and reflect annotator sparsity — MTAT labels are what a handful of players happened to enter, not an exhaustive description — more than model error. The cross-attention distribution concentrates on *soloists* and *bach*, the two tokens in the metadata string that actually carry genre information.

Qualitative retrieval examples — ten caption queries with their top-3 retrieved clips — are written to `results/retrieval_exam`

and the DAC Crowell example discussed in Section 7 is representative: near-misses are semantically correct and penalised only by exact-clip scoring.

The per-tag breakdowns (`results/plots/*_per_tag.png`) explain where the fusion gain actually comes from, and it is not where one might guess. The graph branch dominates on frequent tags with a strong global acoustic signature that survives temporal pooling: *rock* (F1 0.800 vs 0.534 for text), *quiet* (0.430 vs 0.172), *heavy* (0.661 vs 0.416), *piano* (0.688 vs 0.504) and *techno* (0.668 vs 0.520). The text branch wins only on *rare* tags, where the acoustic evidence is thin and catalogue metadata is comparatively informative: *folk* (0.272 vs 0.110, support 41), *cello* (0.182 vs 0.080, support 20), *trance* (0.142 vs 0.051, support 59) and *jazz* (0.350 vs 0.281, support 120).

Fusion gains almost exclusively on that long tail. The largest concatenation improvements over the graph branch are *sitar* (0.369 $\rightarrow$ 0.516), *cello* (0.080 $\rightarrow$ 0.261), *jazz* (0.281 $\rightarrow$ 0.416), *bass* (0.078 $\rightarrow$ 0.186) and *trance* (0.051 $\rightarrow$ 0.172) — every one a tag with fewer than 130 positives in the test split. On the frequent tags concatenation is level with or slightly below the graph branch alone. Because Macro-F1 weights all 50 tags equally, these long-tail gains are what lift the aggregate score, and it follows that the text modality’s value here is to rescue tags the audio branch cannot learn from few examples — not to add complementary information across the board.

9 Limitations

The text modality is metadata, not description. As Section 3.2 explains, MTAT has no captions. Everything here about “language understanding” should be read as understanding of catalogue metadata plus long-tail annotation vocabulary. A corpus with genuine natural-language captions (MusicCaps) would test the claim properly; that requires per-clip audio retrieval from YouTube, which is a separate data-collection project.

Emotion regression is implemented but not trained. The multi-task valence/arousal head and its MAE/ R^2 metrics are implemented (`MultiTaskLoss, metrics.emotion_metrics`) and enabled by `model.emotion.enabled`, but they are disabled by default because they require DEAM, a separate corpus with no clip-level correspondence to MTAT. Reporting emotion numbers here would mean inventing an alignment that does not exist.

Chord estimation is template matching. Binary triad templates over chroma are adequate for building transition graphs but are not a state-of-the-art chord recogniser; seventh chords, inversions and modulations are approximated. Errors here propagate into the chord-node and membership edges.

Single seed. Results are from one seed per configuration. Small differences between adjacent rows should not be over-interpreted without a multi-seed variance estimate.

10 Conclusion

We built and evaluated a GNN-BERT system for musical context understanding that represents each clip as a graph of segments and chords, encodes it with GraphSAGE, and fuses it with a BERT encoding of the clip’s textual context through readable cross-attention. The four-task progression — text-only, graph-only, fusion, contrastive retrieval — is reported under one controlled setup with validation-fitted thresholds and an album-disjoint split, and the ablation isolates the contribution of the fusion mechanism itself. The clearest direction for future work is a corpus with real captions, which would let the language branch carry the semantic weight the architecture is designed for.

Reproducibility

Code, configuration and instructions accompany this report in the submitted `gnn-bert-music-context/` repository. Every hyperparameter is in `config.yaml`; `scripts/run_all.py` reproduces every number in this report, and `tests/test_smoke.py` verifies the full pipeline on synthetic audio without any download.

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